Date: 15-10-2026

CO-1 - AT-3

CODING CHALLENGE

COURES CODE: DSA0308

COUSRE NAME: NLP

Reg no: 192472313 **Name: Pavan Jana**

Section 1: Intelligent Email and Password Validator

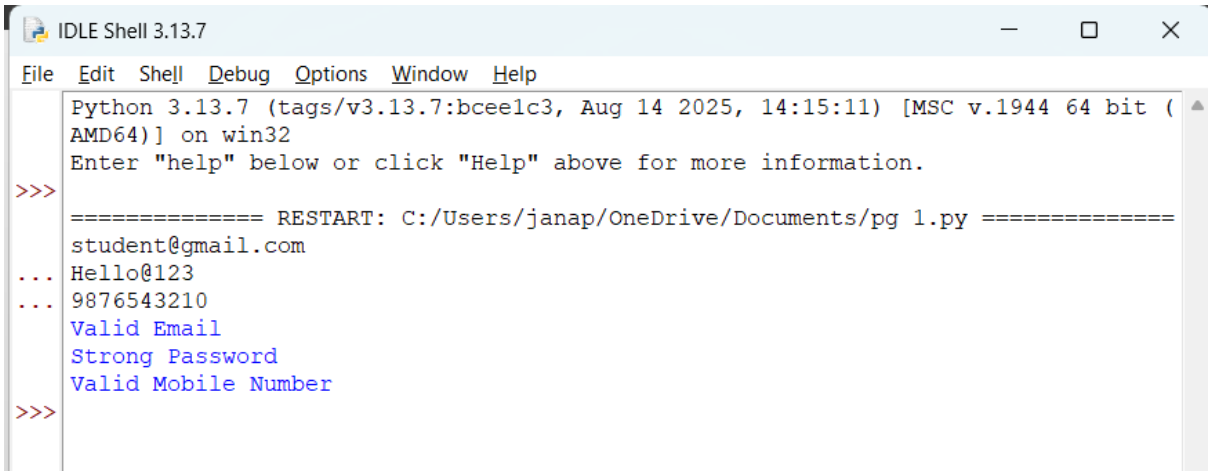
Program:

```
import re
email = input()
password = input()
mobile = input()
if re.fullmatch(r'[A-Za-z][A-Za-z0-9._]*@[A-Za-z]+\.(com|org|edu|net|in)', email):
    print("Valid Email")
else:
    print("Invalid Email")

if re.fullmatch(r'(?=.*[A-Z])(?=.*[a-z])(?=.*\d)(?=.*[@#$%&!]).{8,}', password):
    print("Strong Password")
else:
    print("Weak Password")

if re.fullmatch(r'[6-9]\d{9}', mobile):
    print("Valid Mobile Number")
else:
    print("Invalid Mobile Number")
```

output:



Section 2: Finite State Automata (FSA) Simulator

Program:

```
states = input().split()
alphabet = input().split()
n = int(input())
transitions = {}
for i in range(n):
    state, symbol, next_state = input().split()
    transitions[(state, symbol)] = next_state
initial = input()
final_states = input().split()
count = int(input())

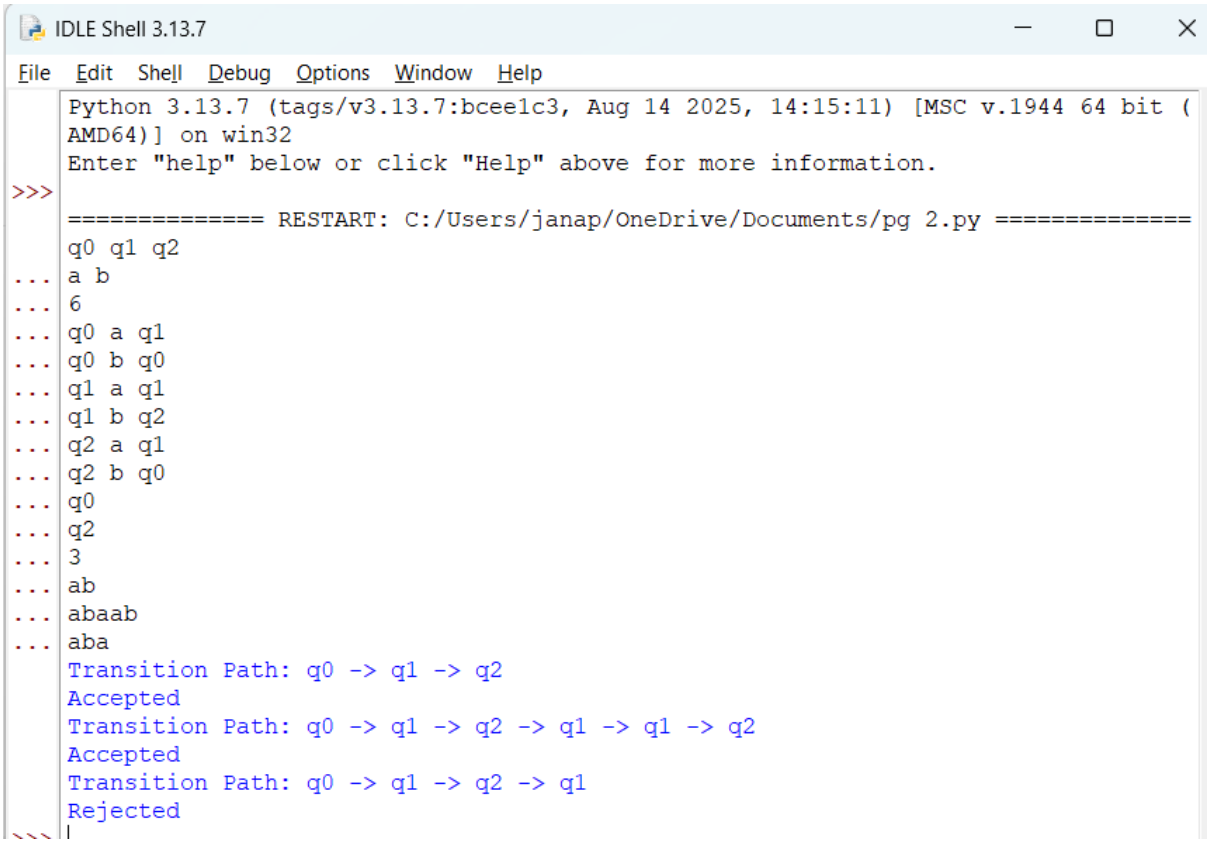

for i in range(count):
    string = input()
    current = initial
    path = [current]
    valid = True
    for symbol in string:
```

```
        if symbol not in alphabet or (current, symbol) not in transitions:
            valid = False
            break
        current = transitions[(current, symbol)]
        path.append(current)

    print("Transition Path:", " -> ".join(path))

    if valid and current in final_states:
        print("Accepted")
    else:
        print("Rejected")
```

OUTPUT:



Section 3: Smart Pattern Matching Engine

Program:

```
import re
text = input()
print("1. Search Date")
print("2. Search Phone Number")
print("3. Search Hashtag")
print("4. Search Mention")
print("5. Search Prefix")
print("6. Search Suffix")
print("7. Search Word")
choice = input()

if choice == "1":
    pattern = r'\b\d{2}/\d{2}/\d{4}\b'
elif choice == "2":
    pattern = r'\b[6-9]\d{9}\b'
elif choice == "3":
    pattern = r'#[A-Za-z0-9_]+'
elif choice == "4":
    pattern = r'@[A-Za-z0-9_]+'
elif choice in ["5", "6", "7"]:
    word = input()

    if choice == "5":
        pattern = r'\b' + re.escape(word) + r'\w*\b'
    elif choice == "6":
        pattern = r'\b\w*' + re.escape(word) + r'\b'
    else:
```

```
        pattern = r'\b' + re.escape(word) + r'\b'
else:
    pattern = None
if pattern:
    result = re.findall(pattern, text, re.IGNORECASE)
    if result:
        print("Matching Patterns:", ", ".join(result))
    else:
        print("No Match Found")
else:
    print("Invalid Choice")
```

OUTPUT:

